RK806 Developer Guide

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Preface

Overview

This document primarily introduces the various sub-modules of the RK806, covering related concepts, functionalities, DTS configuration, and analysis and troubleshooting of common issues.

Product Version

Chip Models	Kernel Version
RK806	5.10

Target Audience

This document (this guide) is primarily intended for the following engineers:

- Technical Support Engineers
- Software Development Engineers

Revision History

Version	Author	Date	Description
V1.0.0	Xu Shengfei	2021-12-24	Initial version

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1. Basics

1.1 Overview

The RK806 is a high-performance PMIC integrating 10 high-current DCDC converters, 5 NLDOs, 6 PLDOs, and configurable power-on sequencing.

Power rails in the system are generally categorized into two types: DCDC and LDO. The general characteristics of these two power types are as follows (please search for detailed information independently):

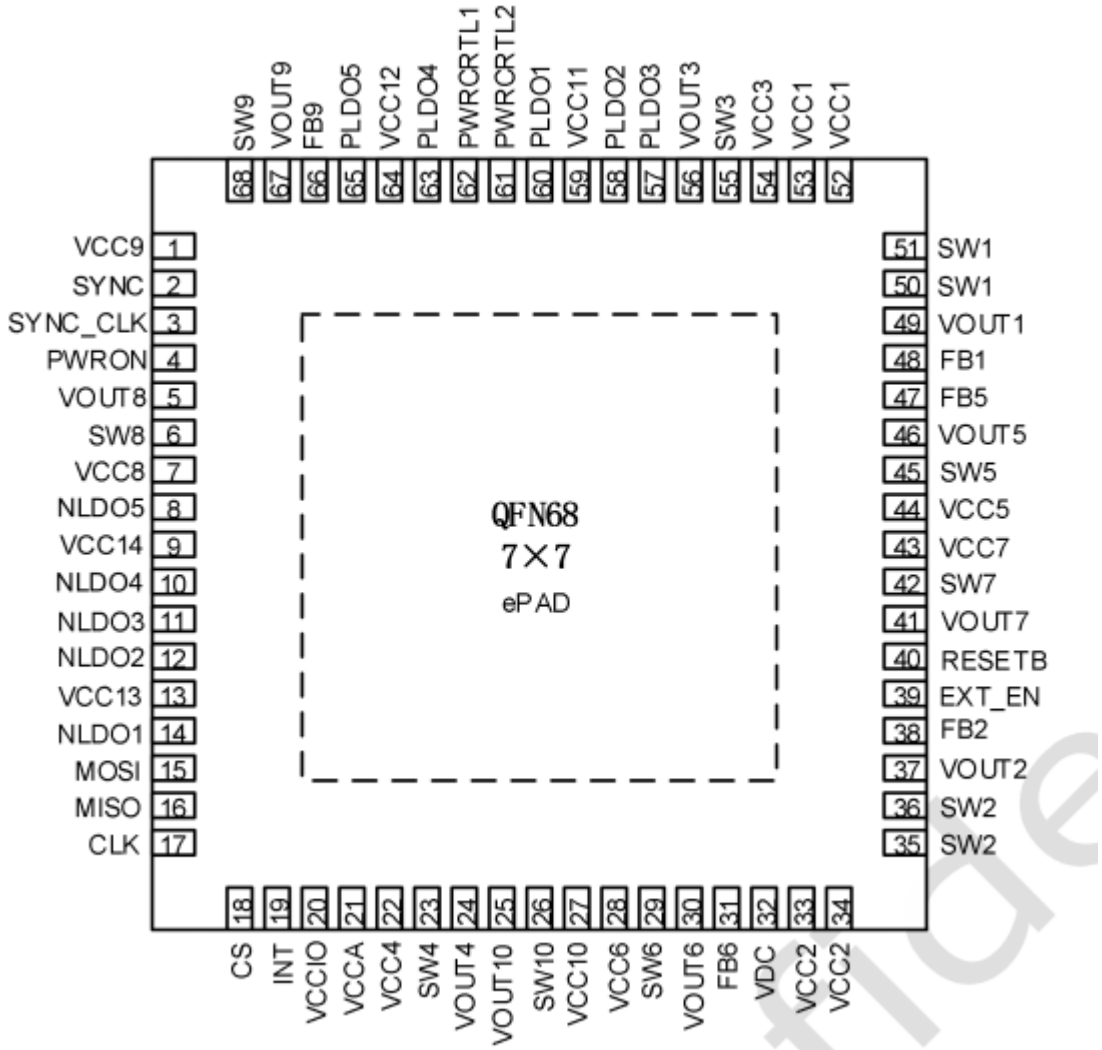
1. DCDC: Offers high efficiency when there is a large difference between input and output voltage, but suffers from relatively large ripple and higher cost. Therefore, it is used for applications with large voltage differentials and high current loads. It generally operates in two modes. PWM mode: Good ripple and transient response, but lower efficiency; PFM mode: High efficiency, but poor load capability.
2. PLDO: Same as LDOs in previous PMICs (minimum input voltage 2.0V, maximum 5.5V).
3. NLDO: Supports low-voltage input and output with lower loss (e.g., 1.1V input to 0.9V output).

1.2 Functions

From the user's perspective, the functions of the RK806 can be summarized into three parts:

1. Regulator function: Controls the status of each DCDC and LDO power rail;
2. GPIO function: Provides 3 available IOs, which can control the entire PMIC to enter standby, assign specific power rails to enter standby individually, or be used as general-purpose GPIOs;
3. PWRKEY function: Detects power button press/release events, saving one GPIO for the AP.

1.3 Chip Pin Functions



In the following description, special attention should be paid to the SLEEP and INT pins:

PIN NO	PIN NAME	PIN DESCRIPTION	I/O
1	VCC9	Power supply of buck9.	I
2	SYNC	Master and slave synchronization signal.	I/O
3	SYNC_CLK	32k synchronization clk.	I/O
4	PWRON	Power on key. The internal pull-up resistance is about 45K.	I
5	VOUT8	Output feedback voltage of buck8.	O
6	SW8	Switching node of buck8.	O
7	VCC8	Power supply of buck8.	I
8	NLDO5	NMOS LDO5 output.	O
9	VCC14	Power supply of NLDO3/4/5.	I
10	NLDO4	NMOS LDO4 output.	O
11	NLDO3	NMOS LDO3 output.	O
12	NLDO2	NMOS LDO2 output.	O
13	VCC13	Power supply of NLDO1/2.	I
14	NLDO1	NMOS LDO1 output.	O
15	MOSI/SCL	SPI MOSI. I2C SCL.	I/O
16	MISO/SDA/ PWRCRTL3	SPI MISO. I2C SDA. PWRCRTL3 control.	I/O
17	CLK	SPI CLK.	O
18	CS	CS is used to select the I2C and SPI functions, when it connected to VCCA is the I2C mode	I
19	INT	Interruput.	O
20	VCCIO/ PLDO6	PMOS LDO6 output. VCCIO for SPI/I2C interface.	O
21	VCCA	Analog power supply. Power supply of PLDO6 and system	I

PIN NO	PIN NAME	PIN DESCRIPTION	I/O
		logic.	
22	VCC4	Power supply of buck4.	I
23	SW4	Switching node of buck4.	O
24	VOUT4	Output feedback voltage of buck4.	O
25	VOUT10	Output feedback voltage of buck10.	O
26	SW10	Switching node of buck10.	O
27	VCC10	Power supply of buck10.	I
28	VCC6	Power supply of buck6.	I
29	SW6	Switching node of buck6.	O
30	VOUT6	Output feedback voltage of buck6.	O
31	FB6	Externed divided resistor mode feedback voltage of buck6.	O
32	VDC	VDC power on signal.	I
33	VCC2	Power supply of buck2.	I
34	VCC2	Power supply of buck2.	I
35	SW2	Switching node of buck2.	O
36	SW2	Switching node of buck2.	O
37	VOUT2	Output feedback voltage of buck2.	O
38	FB2	Externed divided resistor mode feedback voltage of buck2.	O
39	EXT_EN	Control externed DCDC enable. Master/Slave select.	I
40	RESETB	Reset the AP.	I/O
41	VOUT7	Output feedback voltage of buck7.	O
42	SW7	Switching node of buck7.	O
43	VCC7	Power supply of buck7.	I
44	VCC5	Power supply of buck5.	I
45	SW5	Switching node of buck5.	O
46	VOUT5	Output feedback voltage of buck5.	O
47	FB5	Externed divided resistor mode feedback voltage of buck5.	O
48	FB1	Externed divided resistor mode feedback voltage of buck1.	O
49	VOUT1	Output feedback voltage of buck1.	O
50	SW1	Switching node of buck1.	O
51	SW1	Switching node of buck1.	O
52	VCC1	Power supply of buck1.	I
53	VCC1	Power supply of buck1.	I
54	VCC3	Power supply of buck3.	I
55	SW3	Switching node of buck3.	O
56	VOUT3	Output feedback voltage of buck3.	O
57	PLDO3	PMOS LDO3 output.	O
58	PLDO2	PMOS LDO2 output.	O
59	VCC11	Power supply of PLDO1/2/3.	I
60	PLDO1	PMOS LDO1 output.	O
61	PWRCRTL2	PWRCRTL2 control.	I/O
62	PWRCRTL1	PWRCRTL1 control.	I/O
63	PLDO4	PMOS LDO4 output.	O
64	VCC12	Power supply of PLDO4/5.	I
65	PLDO5	PMOS LDO5 output.	O
66	FB9	Externed divided resistor mode feedback voltage of buck9.	O
67	VOUT9	Output feedback voltage of buck9.	O
68	SW9	Switching node of buck9.	O
Exposed	ePAD	Ground	

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1.4 Key Concepts

- Supports SPI and I2C communication

Currently, only SPI communication mode is supported in software, with a default SPI frequency of 1 MHz.

- PMIC has three operating modes

1. PMIC Normal Mode

When the system is running normally, the PMIC operates in normal mode. In this state, the active level of `pmic_sleep` is configurable.

2. PMIC Sleep Mode

To minimize standby power consumption during system suspend, the PMIC switches to sleep mode to reduce its own power usage. Typically, this involves lowering the output voltage of certain rails or disabling outputs entirely, depending on specific product requirements. During system standby, the AP configures `pmic_sleep` to sleep mode via an SPI command, then drives the `pmic_sleep` signal high to enter the sleep state. Upon SoC wakeup, `pmic_sleep` returns to low level, causing the PMIC to exit sleep mode.

3. PMIC Shutdown Mode

When the system initiates the shutdown sequence, the PMIC must perform a complete power-down of the system. The AP configures `pmic_sleep` to shutdown mode via an SPI command, then drives the `pmic_sleep` signal high to enter the shutdown state.

- int pin

Normally high; transitions to low when an interrupt occurs. If the Interrupt remains unhandled, the pin stays low.

- pwrctrl1/pwrctrl2/pwrctrl3 pins

These pins can be used as general-purpose GPIOs.

- pwron pin

The pwrkey function requires the hardware power button to be connected to this pin. The Driver uses this pin to detect press and release events.

- Operating modes of individual DCDC rails

  DCDC supports PWM (also known as force PWM) and PFM modes. However, the PMIC features a mode that dynamically switches between PWM and PFM, commonly referred to as AUTO mode. The PMIC supports two operating modes: PWM and AUTO PWM/PFM. While AUTO mode offers higher efficiency, it exhibits poorer ripple and transient response. For system stability, the device is configured to operate in PWM mode during runtime, and switches to AUTO PWM/PFM mode when the system enters sleep.
- PLDO6 Voltage Regulation

The PMIC supplies VCCIO to the AP. These I/O pins must be powered whenever in use, with a supply voltage of at least 1.8V. Therefore, in typical applications, PLDO6 must not be set below 1.8V and must remain enabled during standby.

- Runtime Voltage Adjustment Range for DCDC and LDO

1. Discontinuous DCDC Voltage Range:

Voltage Range (V)	Step Size (mV)	Specific Voltage Levels (V)
0.7125 ~ 1.45	6.25	0.5, 0.50625, ..., 1.5
1.5 ~ 3.4	25	1.5, 1.525, ..., 3.4

2. Continuous NLDO/PLDO Voltage Range:

Voltage Range (V)	Step Size (mV)	Specific Voltage Levels (V)
0.5 ~ 3.4	12.5	0.5125, 0.525, ..., 3.4

1.5 Dual PMIC Cooperative Operation

- Master-Slave: When operating collaboratively, the two PMICs function in a master-slave configuration. The role of each chip is determined by the logic level of the EXT_EN pin during the initial power-up. The chip with EXT_EN shorted to VCCA operates as the Slave, while the chip with EXT_EN left floating or pulled down via a resistor operates as the Master.
- Synchronization: For dual RK806 collaborative operation, the SYNC_CLK and SYNC pins of both chips are interconnected. The Master chip provides the clock signal (SYNC_CLK, frequency approximately 32 kHz) which is received by the Slave chip. The SYNC pin provides the synchronization signal, generating synchronization pulses to manage power-on, power-off, reset, and power sequencing.
- PWRON and RESETB Signals: The PWRON and RESETB signals of the Master and Slave are shorted together respectively. These are used for PMIC power-on enablement and as external reset signal inputs when connected to an external reset button.
- VDC Signal: The VDC pins of the Master and Slave can be shorted together. Alternatively, the VDC pin of the Slave can be connected to the EXT_EN pin of the Master.

2. Configuration

2.1 Driver and menuconfig

RK806 Driver file:

```
drivers/mfd/rk806-core.c
drivers/mfd/rk806-spi.c
drivers/pinctrl/pinctrl-rk806.c
drivers/regulator/rk806-regulator.c
```

Corresponding macro configuration in menuconfig:

```
CONFIG_PINCTRL_RK806=y
CONFIG_MFD_RK806_SPI=y
CONFIG_REGULATOR_RK806=y
```

2.2 DTS Configuration

5.10 Kernel Configuration

DTS configuration includes SPI mounting, GPIO, regulator, and other sections.

Single RK806 DTS configuration

```
&pinctrl {
         pmic {
                soc_slppin_gpio: soc_slppin_gpio {
                        rockchip,pins = <0 RK_PA2
                                RK_FUNC_GPIO &pcfg_output_low>;
                };
                soc_slppin_shutdown: soc_slppin_shutdown {
                        rockchip,pins = <0 RK_PA2
                                RK_FUNC_GPIO &pcfg_output_high>;
                };
        };
};

&spi2 {
        status = "okay";
        assigned-clocks = <&cru CLK_SPI2>;
        assigned-clock-rates = <200000000>;
        num-cs = <&spi2 {
        status = "okay";
        assigned-clocks = <&cru CLK_SPI2>;
        assigned-clock-rates = <200000000>;
        num-cs = <2>;

        rk806single@0 {
                compatible = "rockchip,rk806";
                spi-max-frequency = <1000000>;
                reg = <0x0>;

                interrupt-parent = <&gpio0>;
                interrupts = <7 IRQ_TYPE_LEVEL_LOW>;

                pinctrl-names = "default", "pmic-sleep", "pmic-power-off", "pmic-reset";
                pinctrl-0 = <&soc_slppin_gpio>, <&rk806_dvs1_null>, <&rk806_dvs2_null>,
<&rk806_dvs3_null>;
                pinctrl-1 = <&soc_slppin_gpio>, <&rk806_dvs1_slp>, <&rk806_dvs2_null>,
<&rk806_dvs3_null>;
                pinctrl-2 = <&rk806_dvs1_pwrdn>, <&rk806_dvs2_null>, <&rk806_dvs3_null>;
                pinctrl-3 = <&rk806_dvs1_rst>, <&rk806_dvs2_null>, <&rk806_dvs3_null>;
```

```
/* 2800mv-3500mv */
low_voltage_threshold = <3000>;
/* 2700mv-3400mv */
shutdown_voltage_threshold = <2700>;
/* 140 160 */
shutdown_temperture_threshold = <160>;
hotdie_temperture_threshold = <115>;

/* 0: restart PMU;
 * 1: reset all the power off reset registers,
 *    forcing the state to switch to ACTIVE mode;
 * 2: Reset all the power off reset registers,
 *    forcing the state to switch to ACTIVE mode,
 *    and simultaneously pull down the RESETB PIN for 5mS before releasing
 */
pmic-reset-func = <1>;

vcc1-supply = <&vcc5v0_sys>;
vcc2-supply = <&vcc5v0_sys>;
vcc3-supply = <&vcc5v0_sys>;
vcc4-supply = <&vcc5v0_sys>;
vcc5-supply = <&vcc5v0_sys>;
vcc6-supply = <&vcc5v0_sys>;
vcc7-supply = <&vcc5v0_sys>;
vcc8-supply = <&vcc5v0_sys>;
vcc9-supply = <&vcc5v0_sys>;
vcc10-supply = <&vcc5v0_sys>;
vcc11-supply = <&vcc_2v0_pldo_s3>;
vcc12-supply = <&vcc5v0_sys>;
vcc13-supply = <&vcc_1v1_nldo_s3>;
vcc14-supply = <&vcc_1v1_nldo_s3>;
vcca-supply = <&vcc5v0_sys>;

regulators {
        vdd_gpu_s0: vdd_gpu_mem_s0: DCDC_REG1 {
                regulator-always-on;
                regulator-boot-on;
                regulator-min-microvolt = <550000>;
                regulator-max-microvolt = <950000>;
                regulator-ramp-delay = <12500>;
                regulator-name = "vdd_gpu_s0";
                regulator-state-mem {
                        regulator-off-in-suspend;
                };
        };

        vdd_cpu_lit_s0: vdd_cpu_lit_mem_s0: DCDC_REG2 {
                regulator-always-on;
                regulator-boot-on;
                regulator-min-microvolt = <550000>;
                regulator-max-microvolt = <950000>;
                regulator-ramp-delay = <12500>;
```

```
                                regulator-name = "vdd_cpu_lit_s0";
                                regulator-state-mem {
                                        regulator-off-in-suspend;
                                };
                        };

                        vdd_log_s0: DCDC_REG3 {
                            ...
                        };

                        vdd_vdenc_s0: vdd_vdenc_mem_s0: DCDC_REG4 {
                            ...
                        };
                            ...
                };
        };
};
```

Dual RK806 DTS configuration:

```
&pinctrl {
    pmic {
            soc_slppin_gpio: soc_slppin_gpio {
                    rockchip,pins = <0 RK_PA2
                            RK_FUNC_GPIO &pcfg_output_low>;
            };
            soc_slppin_shutdown: soc_slppin_shutdown {
                    rockchip,pins = <0 RK_PA2
                            RK_FUNC_GPIO &pcfg_output_high>;
            };
    };
};

&spi2 {
        status = "okay";
        assigned-clocks = <&cru CLK_SPI2>;
        assigned-clock-rates = <200000000>;
        num-cs = <2>;
        rk806master@0 {
                compatible = "rockchip,rk806";
                spi-max-frequency = <1000000>;
                reg = <0x0>;

                interrupt-parent = <&gpio0>;
                interrupts = <7 IRQ_TYPE_LEVEL_LOW>;
                /* 0: restart PMU;
                 * 1: reset all the power off reset registers,
                 *    forcing the state to switch to ACTIVE mode;
                 * 2: Reset all the power off reset registers,
                 *    forcing the state to switch to ACTIVE mode,
                 *    and simultaneously pull down the RESETB PIN for 5mS before releasing
                 */
                pmic-reset-func = <1>;
```

```
        vcc1-supply = <&vcc5v0_sys>;
        vcc2-supply = <&vcc5v0_sys>;
        ...
        vcca-supply = <&vcc5v0_sys>;
        regulators {
                vdd_gpu_s0: DCDC_REG1 {
                        regulator-always-on;
                        regulator-boot-on;
                        regulator-min-microvolt = <550000>;
                        regulator-max-microvolt = <950000>;
                        regulator-ramp-delay = <12500>;
                        regulator-name = "vdd_gpu_s0";
                        regulator-state-mem {
                                regulator-off-in-suspend;
                        };
                };
                vdd_npu_s0: DCDC_REG2 {
                ...
                };
                ...
        };
};
rk806slave@1 {
        compatible = "rockchip,rk806";
        spi-max-frequency = <1000000>;
        reg = <0x01>;

        interrupt-parent = <&gpio0>;
        interrupts = <7 IRQ_TYPE_LEVEL_LOW>;
        /* 0: restart PMU;
         * 1: reset all the power off reset registers,
         *    forcing the state to switch to ACTIVE mode;
         * 2: Reset all the power off reset registers,
         *    forcing the state to switch to ACTIVE mode,
         *    and simultaneously pull down the RESETB PIN for 5mS before releasing
         */
        pmic-reset-func = <1>;

        vcc1-supply = <&vcc5v0_sys>;
        vcc2-supply = <&vcc5v0_sys>;
        ...
        vcca-supply = <&vcc5v0_sys>;

        pwrkey {
                status = "disabled";
        };
        regulators {
                vdd_cpu_big1_s0: DCDC_REG1 {
                        regulator-always-on;
                        regulator-boot-on;
                        regulator-min-microvolt = <550000>;
                        regulator-max-microvolt = <950000>;
```

```
                                regulator-ramp-delay = <12500>;
                                regulator-name = "vdd_cpu_big1_s0";
                                regulator-state-mem {
                                        regulator-off-in-suspend;
                                };
                        };
                        vdd_cpu_big0_s0: DCDC_REG2 {
                        ...
                        };
                        ...
                };
        };
};
```

1. SPI Mounting

The complete RK806 node must be attached under the corresponding SPI node, with status configured as "okay".

2. Main Section

- Do not modify:

```
compatible = "rockchip,rk806";
spi-max-frequency = <1000000>;
reg = <0x0>;

interrupt-parent = <&gpio0>;
interrupts = <7 IRQ_TYPE_LEVEL_LOW>;
```

- Modifiable (according to pinctrl rules)

interrupt-parent: Specifies which GPIO the pmic_int belongs to. interrupts: Specifies the pin index and polarity of pmic_int on the GPIO defined in interrupt-parent.

3. pwrkey, gpio

If the pwrkey or GPIO functionality is not used in the project, add the pwrkey or GPIO node in the DTS and explicitly set the status to `status = "disabled"`. This prevents the Driver from being enabled; however, error logs may appear during boot and can be ignored.

```
pwrkey {
        status = "disabled";
};
```

4. regulator

- `regulator-compatible`: The name used for matching during Driver registration. Do not modify this value, otherwise the Driver will fail to load.
- `regulator-name`: The power supply name. It is recommended to keep this consistent with the hardware schematic. This name must be matched when using the `regulator_get` interface.
- `regulator-init-microvolt`: The initialization voltage during the U-Boot stage; this is invalid in the kernel stage.

- `regulator-min-microvolt`: The minimum adjustable voltage during runtime.
- `regulator-max-microvolt`: The maximum adjustable voltage during runtime.
- `regulator-initial-mode`: The DCDC operating mode during runtime, typically configured as 1. 1: force PWM, 2: auto PWM/PFM.
- `regulator-mode`: The DCDC operating mode during sleep, typically configured as 2. 1: force PWM, 2: auto PWM/PFM.
- `regulator-initial-state`: The mode during suspend; must be configured as 3.
- `regulator-boot-on`: If this property exists, the power supply will be enabled upon Driver registration.
- `regulator-always-on`: If this property exists, it indicates that the power supply cannot be disabled during runtime and will be enabled upon Driver registration.
- `regulator-ramp-delay`: The DCDC voltage rise time, fixed at 12500.
- `regulator-on-in-suspend`: Maintains the power-on state during sleep. To disable this power supply during sleep, change this to `regulator-off-in-suspend`.
- `regulator-suspend-microvolt`: The standby voltage when power is maintained during sleep.

5.10 Kernel Configuration

Refer to the 5.10 kernel DTS configuration.

2.3 Function Interfaces

The following interfaces generally meet daily usage requirements, including regulator enable/disable, voltage setting, and voltage retrieval:

1. Get regulator:

   `struct regulator *devm_regulator_get_optional(struct device *dev, const char *id)`

   `dev` refers to the current device, and `id` corresponds to the `-supply` property configured in the DTS.

2. Release regulator

`void regulator_put(struct regulator *regulator)`

3. Enable regulator

`int regulator_enable(struct regulator *regulator)`

4. Disable regulator

`int regulator_disable(struct regulator *regulator)`

5. Get regulator voltage

`int regulator_get_voltage(struct regulator *regulator)`

6. Set regulator voltage

`int regulator_set_voltage(struct regulator *regulator, int min_uV, int max_uV)`

The caller must ensure that the passed parameters satisfy `min_uV <= max_uV`.

7. Example

```
struct regulator *vdd_ana;

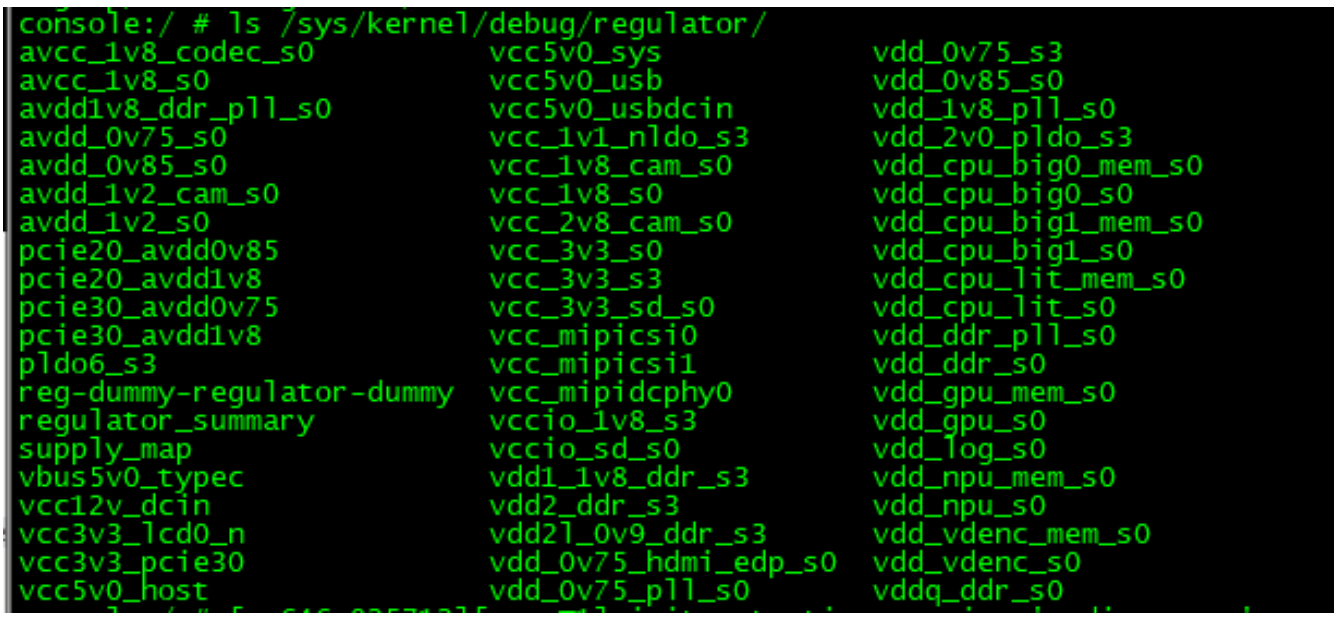
vdd_ana = devm_regulator_get_optional(dev, "power");
/* Obtain power-supply from dts
   power-supply = <&vcc3v3_lcd0_n> */
regulator_enable(vdd_ana);
regulator_disable(vdd_ana);                         // Disable vdd_ana
regulator_put(vdd_ana);                             // Release vdd_ana
```

3. Debug

3.1 Kernel

Since the drivers related to the PMIC are not complex in terms of usage logic, with the focus primarily on the final Register configurations, the commonly used debug methods can be accessed via the following nodes:

`/sys/kernel/debug/regulator`



`/sys/kernel/debug/regulator/regulator_summary`

```
 regulator                     use open bypass  opmode voltage current     min     max
-------------------------------------------------------------------------------------
 regulator-dummy                 5    5      0 unknown     0mV      0mA     0mV     0mV
    fe210000.sata-target         1                                 0mA     0mV     0mV
    fe210000.sata-phy            1                                 0mA     0mV     0mV
    fe210000.sata-ahci           1                                 0mA     0mV     0mV
    backlight-power              1                                 0mA     0mV     0mV
    regulator-dummy              0                                 0mA     0mV     0mV
 vcc12v_dcin                     3    4      0 unknown 12000mV      0mA 12000mV 12000mV
    vcc12v_dcin                  0                                 0mA     0mV     0mV
    vcc5v0_sys                  30   30      0 unknown  5000mV      0mA  5000mV  5000mV
       vcc5v0_sys                0                                 0mA     0mV     0mV
       vdd_gpu_s0                1    3      0  normal   750mV      0mA   550mV   950mV
          fb000000.gpu-mali      0                                 0mA   750mV   950mV
          fb000000.gpu-mali      0                                 0mA     0mV     0mV
          vdd_gpu_s0             0                                 0mA     0mV     0mV
       vdd_npu_s0                1    1      0  normal   750mV      0mA   550mV   950mV
          vdd_npu_s0             0                                 0mA     0mV     0mV
       vdd_log_s0                1    1      0  normal   750mV      0mA   750mV   750mV
          vdd_log_s0             0                                 0mA     0mV     0mV
       vdd_vdenc_s0              1    1      0  normal   750mV      0mA   550mV   950mV
          vdd_vdenc_s0           0                                 0mA     0mV     0mV
       vdd_gpu_mem_s0            1    3      0  normal   750mV      0mA   675mV   950mV
          fb000000.gpu-mem       0                                 0mA   750mV   950mV
          fb000000.gpu-mem       0                                 0mA     0mV     0mV
          vdd_gpu_mem_s0         0                                 0mA     0mV     0mV
       vdd_npu_mem_s0            1    1      0  normal   750mV      0mA   675mV   950mV
          vdd_npu_mem_s0         0                                 0mA     0mV     0mV
       vdd_2v0_pldo_s3           9    9      0  normal  2000mV      0mA  2000mV  2000mV
          vdd_2v0_pldo_s3        0                                 0mA     0mV     0mV
          avcc_1v8_s0            1    3      0 unknown  1800mV      0mA  1800mV  1800mV
             avcc_1v8_s0         0                                 0mA     0mV     0mV
             pcie20_avdd1v8      0    1      0 unknown  1800mV      0mA  1800mV  1800mV
                pcie20_avdd1v8   0                                 0mA     0mV     0mV
             pcie30_avdd1v8      0    1      0 unknown  1800mV      0mA  1800mV  1800mV
                pcie30_avdd1v8   0                                 0mA     0mV     0mV
          vdd1_1v8_ddr_s3        1    1      0 unknown  1800mV      0mA  1800mV  1800mV
             vdd1_1v8_ddr_s3     0                                 0mA     0mV     0mV
          avcc_1v8_codec_s0      1    1      0 unknown  1800mV      0mA  1800mV  1800mV
             avcc_1v8_codec_s0   0                                 0mA     0mV     0mV
          avdd_1v2_cam_s0        1    1      0 unknown  1200mV      0mA  1200mV  1200mV
             avdd_1v2_cam_s0     0                                 0mA     0mV     0mV
          avdd_1v2_s0            1    1      0 unknown  1200mV      0mA  1200mV  1200mV
             avdd_1v2_s0         0                                 0mA     0mV     0mV
          vcc_1v8_cam_s0         1    1      0 unknown  1800mV      0mA  1800mV  1800mV
             vcc_1v8_cam_s0      0                                 0mA     0mV     0mV
          avdd1v8_ddr_pll_s0     1    1      0 unknown  1800mV      0mA  1800mV  1800mV
             avdd1v8_ddr_pll_s0  0                                 0mA     0mV     0mV
          vdd_1v8_pll_s0         1    1      0 unknown  1800mV      0mA  1800mV  1800mV
             vdd_1v8_pll_s0      0                                 0mA     0mV     0mV
       vdd_vdenc_mem_s0          1    1      0  normal   750mV      0mA   675mV   950mV
          vdd_vdenc_mem_s0       0                                 0mA     0mV     0mV
       vdd2_ddr_s3               1    1      0  normal  1100mV      0mA     0mV     0mV
          vdd2_ddr_s3            0                                 0mA     0mV     0mV
       vcc_1v1_nldo_s3           6    6      0  normal  1100mV      0mA  1100mV  1100mV
          vcc_1v1_nldo_s3        0                                 0mA     0mV     0mV
          avdd_0v75_s0           1    2      0 unknown   750mV      0mA   750mV   750mV
```

The attributes and voltage values for each power rail can be found via the following nodes:

```
console:/sys/kernel/debug/regulator/vdd_gpu_mem_s0 # ls
bypass_count  fb000000.gpu-mem  mode         vdd_gpu_mem_s0
consumers     force_disable     open_count   voltage
enable        load              use_count
```

3.2 Kernel 5.10

Please refer to the Kernel 5.10 commands.